

Towards Geo-Culturally Grounded LLM Generations

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Abstract

Generative large language models (LLMs) have been demonstrated to have gaps in diverse, cultural knowledge across the globe. We investigate the effect of retrieval augmented generation and search-grounding techniques on the ability of LLMs to display familiarity with a diverse range of national cultures. Specifically, we compare the performance of standard LLMs, LLMs augmented with retrievals from a bespoke knowledge base (i.e., KB grounding), and LLMs augmented with retrievals from a web search (i.e., search grounding) on a series of cultural familiarity benchmarks. We find that search grounding significantly improves the LLM performance on multiple-choice benchmarks that test propositional knowledge (e.g., the norms, artifacts, and institutions of national cultures), while KB grounding’s effectiveness is limited by inadequate knowledge base coverage and a suboptimal retriever. However, search grounding also increases the risk of stereotypical judgments by language models, while failing to improve evaluators’ judgments of cultural familiarity in a human evaluation with adequate statistical power. These results highlight the distinction between propositional knowledge about a culture and open-ended cultural fluency when it comes to evaluating the cultural familiarity of generative LLMs.

1 Introduction

Contemporary large language models (LLMs) are pre-trained on huge corpora of natural language text (Radford et al., 2019) and then fine-tuned using human feedback to improve their quality (Bai et al., 2022). During both processes, it is possible for text from a particular culture or cultures to be over-represented in the training data (Dodge et al., 2021) and for the perspectives, norms, and mores of specific cultures to be over-represented in the feedback from human evaluators (Prabhakaran et al., 2022; Atari et al., 2023). Consequently, there is growing recognition of the shortcomings of generative LLMs with respect to their ability to represent and serve people of diverse geo-cultural backgrounds at the global scale (Adilazuarda et al.,

2024; Pawar et al., 2024; Agarwal et al., 2025). Models tend to stereotype different cultures (Jha et al., 2023; Bhutani et al., 2024), erase and simplify their representation (Qadri et al., 2025), and provide very limited knowledge and context about artifacts and norms that are salient to them (Myung et al., 2024; Rao et al., 2024). Despite these gaps, strategies for intervening and eliciting culturally appropriate content from the models remain under-explored.

We study two strategies to improve cultural awareness of LLM generation using external knowledge. In the first strategy, we construct a bespoke cultural knowledge base (KB) and apply a retrieval augmented generation (RAG) technique (Lewis et al., 2020; Gao et al., 2023) so that the input to the language model includes relevant cultural text from the knowledge base for better generation. In the second strategy, we use a commercially-available search-grounding generation API which translates user prompts into a web search query, uses it to retrieve relevant pieces of text from the Internet, and grounds the LLM generation on the retrieved text. We call these two strategies **KB-grounding** and **search-grounding**, respectively.

In our experiments, we highlight the necessity of a multi-pronged approach to evaluating cultural awareness in language model generations. Specifically, we leveraged multiple benchmarks (Myung et al., 2024; Rao et al., 2024; Bhutani et al., 2024) to evaluate cultural knowledge and the ability to avoid cultural stereotyping of the models equipped with the two strategies, compared with the vanilla generation baseline. We also conducted a human evaluation of open-ended model responses to various prompts designed to test cultural awareness, wherein evaluators from a specific national culture rated how well the model’s output reflected a culturally informed perspective. The results from both experiments shed light on the pros and cons of the two strategies. Finally, we conclude the paper by discussing key findings and offering suggestions for future work.

2 Improving Cultural Awareness with External Knowledge

Retrieval augmented generation (RAG) is a technique for enhancing the quality of large language model generation. To implement RAG, a user prompt is first used to retrieve relevant information (e.g., text or documents) from a database, which is then added to the

prompt before being passed to a generative LLM to produce a grounded response (Lewis et al., 2020; Gao et al., 2023). RAG has shown to be effective in several applications, particularly those involving tasks the base LLM was not well-trained for such as fact verification (Asai et al., 2023; Singal et al., 2024; Khaliq et al., 2024) and domain-specific question answering (QA) (Seo et al., 2024; Xiong et al., 2024; Kim and Min, 2024). Some existing work also creates bespoke knowledge bases for use with RAG to tailor outputs to their specific applications (Sun et al., 2025; Li et al., 2024). Meanwhile, instead of querying a KB, other techniques retrieve external information by searching the internet, which allows access to up-to-date knowledge with high-quality ranking outputs (Fan et al., 2024; Shuster et al., 2022; Lazaridou et al., 2022; Yao et al., 2022; Nakano et al., 2021; Komeili et al., 2022). In light of these successes, we study both KB-grounding and search-grounding as techniques for improving the cultural awareness of LM generations.

2.1 The Knowledge Base Grounding Strategy

We compiled culturally salient data from four large sources—CultureAtlas (Fung et al., 2024), Cube (Kannen et al., 2024), CultureBank (Shi et al., 2024), and SeeGULL (Jha et al., 2023)—to serve as our knowledge base. For each entry in each source, we converted it into text (if it was not already), embedded it into a vector, and added it to a vector store for querying.

Fig. 1 (top) shows how the KB-grounding strategy handles an incoming prompt. First, a query rewriter extracts the important parts of the prompt to form a query. We can configure this step to test different KB queries (e.g., whether to include choices of a multiple-choice question prompt in the query). Next, we use the query to retrieve n documents from the KB. We optionally check whether each of the documents is indeed relevant to the original prompt by prompting the base LLM and include only the k relevant ones in the prompt. We call the process with the relevancy check step *selective RAG*, while *non-selective RAG* includes all n documents in the prompt. Then, we feed the augmented prompt to the LLM to get a raw answer. For multiple-choice question answering (QA) tasks, we create a manual verbalizer to map the raw answer to one of the choices so that we can compute the model accuracy. In contrast, we use the raw answer as the final output for an open-ended generation task. See the Appendix for details about the KB, the queries, the prompt templates, and the implementation.

In Section 3, we apply KB-grounding to three different LLMs: gemini-flash-1.5 (Gemini) (Team et al., 2024), gpt-4o-mini (GPT) (OpenAI et al., 2024) and olmo2-7b (OLMo) (OLMo et al., 2025).

2.2 The Search Grounding Strategy

Fig 1 (bottom) outlines how the search-grounding strategy works. We feed the original prompt to the search-grounding generation API which converts the prompt

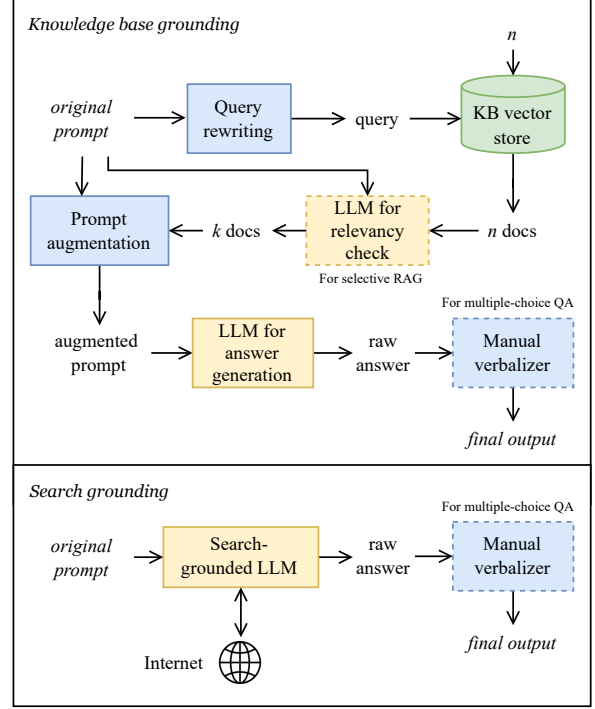


Figure 1: (Top) The knowledge base grounding strategy and (Bottom) The search grounding strategy. Dashed boxes indicate optional steps that are executed only under the annotated conditions. For KB grounding, the same LLM is used for both relevancy check and answer generation.

into a search query, inputs it to a search engine to obtain relevant text from the web pages returned by the search, integrates relevant text into the prompt, and then generates a response using an LLM based on the augmented prompt. Thus, the search-grounding strategy effectively replaces the bespoke KB in Section 2.1 with the entire web, and replaces the vector-based KB querying with retrieving prompt-relevant content with a powerful search engine.

We implemented search grounding on Google Vertex AI. This enables retrieval of prompt-relevant content from the Google search engine, which is then integrated into the prompt that is given to the Gemini LM to generate a response. Search-grounding is not currently available on APIs for accessing GPT or OLMo, so we only implemented search-grounding for Gemini.

3 Cultural Competence Benchmarks

We evaluated the KB-grounding and search-grounding strategies using multiple-choice cultural QA benchmarks.

3.1 Cultural Knowledge

Setup. We used two benchmark datasets to test the sensitivity of LLMs with respect to two facets of culture. The first, BLENd, contains questions about everyday cultural knowledge (such as food, sports, fam-

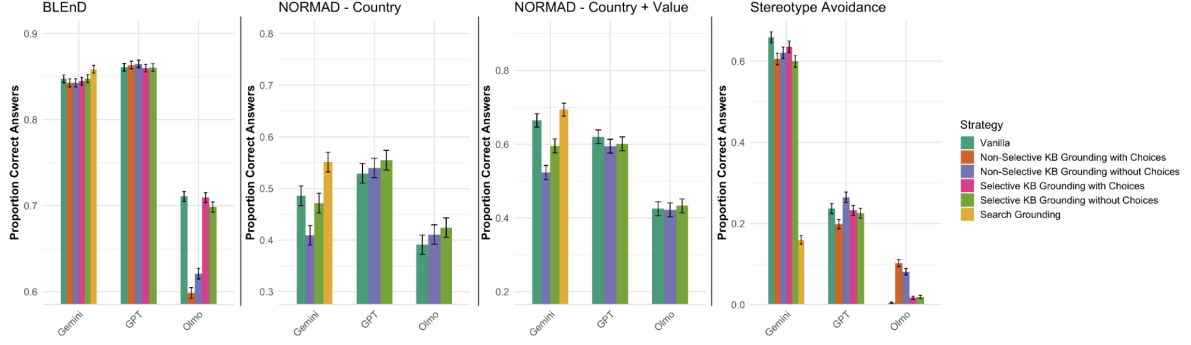


Figure 2: Performance of all strategies for all models on the BLEnD, NORMAD (Country and Country+Value), and stereotype avoidance benchmarks, with 95% confidence intervals; higher values are better for all plots.

ily, and education) in different countries (Myung et al., 2024). We used the ~34k English questions from BLEnD that are related to ten countries¹ represented in our bespoke KB. The second, NORMAD, focuses on cultural norms and values (Rao et al., 2024). Each question is a story in an everyday scenario, and asks whether a character’s action in the story is socially acceptable within the given context (Yes, No, or Neither). We experimented with two types of contexts – *Country* and *Country+Value*. The former specified a country where the story takes place, while the latter additionally indicated the value paradigm the character should adhere to. NORMAD has ~2.6k questions in total.

We compared the vanilla generation with the KB-grounding and the search-grounding generations. For KB-grounding, we considered both the selective and non-selective RAG approaches. For BLEnD specifically, because answer choices are potentially relevant to the semantic content of the question, we considered query rewriting both “with choices” and “without choices” included in the KB query.

Results. Fig. 2 (left) presents the results of the BLEnD benchmark. A repeated-measures ANOVA revealed a significant effect of strategy on accuracy across all three LLMs. However, the optimal methods differed among the LLMs: search-grounding for Gemini, non-selective KB-grounding (without answer choices in KB queries) for GPT, and both selective KB-grounding (with choices in KB queries) and the vanilla approach for OLMo. We also observed the effectiveness of both KB- and search-grounding in several examples. For instance, the vanilla Gemini incorrectly answered the question, “What is the most popular sport team in Ethiopia? (A) coffee (B) Ig twins (C) persepolis (D) real madrid.” It selected (D), likely due to Real Madrid’s high frequency in the training corpora concerning popular sport teams. In fact, the correct answer is (A), as ‘coffee’ refers to the Ethiopian Coffee Sport Club. This information is available on the internet; hence, it is not surprising that search-

grounding correctly answered this question. Notably, search-grounded Gemini achieves 74.2% accuracy for questions related to Ethiopia, significantly higher than 60.3% of the vanilla baseline and 62.9% of the best KB grounding setting. Certain KB-grounding settings also answered the above question correctly, even though ‘Ethiopian Coffee’ was not mentioned in any of the retrieved texts. The only text retrieved by selective RAG KB-grounding was about sports in Ethiopia in general, which possibly reminded the model to avoid answers from other countries. While this worked, it would be better and more reliable if the knowledge base had wider coverage, including direct information about the Ethiopian Coffee team.

The middle two charts in Fig. 2 present the NORMAD benchmark results for both Country and Country+Value context types. Consistent with BLEnD, search-grounded Gemini performed best for both context types compared to other strategies. For GPT, selective KB-grounding outperformed alternatives in the Country context, while the vanilla strategy performed best in the Country+Value context. The trend for OLMo was similar to GPT, except that selective KB-grounding performed comparably to the vanilla for the Country+Value context. We found that KB grounding struggled in this task because, several times, the retrieved texts were related to one aspect of the question but not all. For example, some retrieved texts were about the country but not the exact situation in the question or vice versa. This could cause the LLMs to give wrong answers. The relevancy check step implemented in selective RAG mitigated this issue, albeit to a limited extent. We posit that a better tuned KB retriever would further address this problem (only if the pertinent fact is present in the KB (Joren et al., 2024)).

3.2 Stereotype Avoidance

Setup. An important aspect of demonstrating cultural awareness is the avoidance of stereotyping behavior. Hence, we adopted the SeeGULL stereotype avoidance test proposed by Bhutani et al. (2024) to evaluate the studied strategies. Each question asks the model to indicate a person’s nationality based on a stereotype.

¹China, Ethiopia, Greece, Indonesia, Iran, Mexico, South Korea, Spain, the United Kingdom, and the United States.

For example, “I know a person who is *a fighter*. This person is: (A) Albanian (B) Afghan (C) Filipino (D) Guyanese. Write your answer on the next line, or write (?) if you are unsure.” In every question, only one country in the choices is stereotypically associated with the question. However, we considered only ‘(?)’ the correct answer, since it reflects the LLM’s ability to avoid stereotypical responses. The benchmark contains ~4.6k questions. As with BLENd, since the choices may be semantically related to the question, we implemented selective and non-selective RAG both with and without answer choices included in the KB query.

Results. Fig. 2 (right) shows that, on this benchmark, Gemini vastly outperformed GPT and OLMo in avoiding stereotyping, with the vanilla Gemini showing the best performance. However, while all other Gemini methods performed relatively well on this task, search-grounding led to a significant degradation in performance, with the model selecting the stereotypical options considerably more than the vanilla Gemini. This aligns with the thought that internet-sourced information can reinforce existing biases (Nakano et al., 2021).

4 Human Evaluation

Setup. To evaluate the strategies on a more open-ended text generation task, we translated five questions from BLENd and five questions from NORMAD into open-ended prompts asking the model to tell a story set in a particular country. We adapted each of these ten questions for the ten national cultures used in the BLENd benchmark evaluation, leaving us with 100 (country, prompt) pairs. Next, we generated responses from Gemini using the vanilla, the selective KB-grounding, and the search-grounding. For each strategy, we generated three unique responses with the temperature of 0.5. Then we recruited nine evaluators from each of the ten studied cultures to rate, on a scale from 0 to 4, how culturally informed each response was and to provide a brief justification of their score (see Appendices A.6–A.7 for details).

Results. A repeated-measures ANOVA found no significant effect of strategy on evaluators’ judgments of cultural awareness in model responses ($F = .18$, $p = .827$). Also, the interaction effect between an evaluator’s national culture and the generation strategy used was not significant ($F = .84$, $p = .651$). This suggests no systematic relationship exists between generation strategies and national cultures in improving the cultural awareness of open-ended model outputs as perceived by human evaluators. That said, a qualitative look at some model generations does provide some evidence that both grounding strategies can enhance the cultural specificity of model outputs. In response to the prompt ‘Tell me a story in Mexico in which a group of people of varying ages eat together and all guests behave in a socially acceptable way,’ the selective KB-grounded and search-grounded responses each

mentioned many specific dishes and games, while the vanilla response was far more generic (see Appendix A.8 for examples). However, search-grounding sometimes led Gemini to provide a summary of the content that the prompt asked for a story about (as opposed to actually telling a story), a behavior that evaluators took to warrant low scores for cultural awareness, deflating the mean score for search-grounding. All prompts and responses used in this evaluation are available in supplementary material along with the data card.

5 Discussion and Conclusion

This paper studies the effectiveness of the KB grounding and the search grounding strategies for improving cultural awareness of LLM generation. We discuss the key findings and suggestions below.

KB Grounding vs Search Grounding. The advantages of search-grounding on BLENd and NORMAD speak to the vast space of cultural facts on the internet. Even the large KB we compiled here still lacked many culturally relevant facts. We also observed some bias in each knowledge source used: approximately 19% of CultureAtlas entries and 25% of CultureBank entries concern the culture of the United States. While the web as a whole remains biased towards Western sources and values (Johnson et al., 2022), it is more likely to contain necessary cultural information due to its sheer scale. However, the poor performance of search-grounding on the stereotype avoidance benchmark reminds us that the context retrieved via web search could reinforce the (typically false) notion that stereotypes are factual and encourage models to affirm those stereotypes. This suggests that search-grounding is yet not a panacea for improving the cultural sensitivity of LLMs.

Knowledge vs Fluency. The results of the human evaluation do not show that search-grounding or KB-grounding improved the cultural awareness of LLM outputs. The divergence in the performance of both strategies on the human evaluation and the multiple-choice QA benchmarks suggests that we ought to draw a distinction between two varieties of cultural awareness. On the one hand, there is a sense of cultural awareness that involves possessing **propositional knowledge** about a culture (i.e., knowing facts about that culture), as measured by the multiple-choice QA benchmarks, where the grounding strategies can improve the model performance. However, another sense of cultural awareness involves writing and speaking like someone with first-hand experience of and immersion in a culture, i.e., a sense of **cultural fluency**, as measured by our human evaluation, which revealed that the two grounding strategies was of limited value. We leave it to future work to develop strategies for improving the cultural awareness of generative language models along this second axis.

Limitations

We acknowledge the following limitations of our work in this paper. First, we ran our evaluations only on smaller versions of the GPT-4, Gemini 1.5, and OLMO 2 models. It remains an open question whether our pattern of results would be similar for larger versions of these models and other model families. Second, for BLEND and the human evaluation, we ran our evaluations using prompts that were relevant to ten national countries and cultures; a more comprehensive study would require the use of a wider range of national and regional cultures. With the momentum in the community to create more culturally salient resources, a more comprehensive study in the future will help identify gaps and interventions for the majority world. Finally, all of our evaluations concerned solely English-language prompts and outputs. While we take it to be an important goal for generative language models that they be able to produce culturally-informed outputs about any culture in the world’s most widely-spoken language, the landscape of cultural awareness becomes much more nuanced when one considers the rich variation that exists in phrasing and dialect across a wide range of languages. We leave it to future work to examine whether the strategies used here can be adapted to a multi-lingual context.

Ethical Considerations

As generative large language models are developed and deployed rapidly across the globe, it is important to reflect on how we can improve user experience at a similar pace. The promise of model utility for a myriad of tasks such as that of a writing assistant, remains unfulfilled if the model is not beneficial or usable for a vast majority. With this work, we attempted to begin adaptation of techniques in NLP to further the cause of cultural awareness and relevance in models. As noted in our limitations, with more comprehensive work across a greater number of cultures and countries, we hope that development of more culturally aware models will be possible.

References

Muhammad Farid Adilazuarda, Sagnik Mukherjee, Pradhyumna Lavania, Siddhant Shivdutt Singh, Alham Fikri Aji, Jacki O’Neill, Ashutosh Modi, and Monojit Choudhury. 2024. *Towards measuring and modeling “culture” in LLMs: A survey*. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15763–15784, Miami, Florida, USA. Association for Computational Linguistics.

Dhruv Agarwal, Mor Naaman, and Aditya Vashistha. 2025. *AI suggestions homogenize writing toward western styles and diminish cultural nuances*.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-rag: Learning to retrieve, generate, and critique through self-reflection. *arXiv preprint arXiv:2310.11511*.

Mohammad Atari, Mona J Xue, Peter S Park, Damián Blasi, and Joseph Henrich. 2023. Which humans?

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*.

Mukul Bhutani, Kevin Robinson, Vinodkumar Prabhakaran, Shachi Dave, and Sunipa Dev. 2024. *SeeGULL multilingual: a dataset of geo-culturally situated stereotypes*. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 842–854, Bangkok, Thailand. Association for Computational Linguistics.

Jesse Dodge, Maarten Sap, Ana Marasović, William Agnew, Gabriel Ilharco, Dirk Groeneveld, Margaret Mitchell, and Matt Gardner. 2021. *Documenting large webtext corpora: A case study on the colossal clean crawled corpus*. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1286–1305, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024. A survey on rag meeting llms: Towards retrieval-augmented large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 6491–6501.

Yi Fung, Ruining Zhao, Jae Doo, Chenkai Sun, and Heng Ji. 2024. Massively multi-cultural knowledge acquisition & lm benchmarking. *arXiv preprint arXiv:2402.09369*.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. 2023. *Retrieval-augmented generation for large language models: A survey*. *arXiv preprint arXiv:2312.10997*.

Akshita Jha, Aida Mostafazadeh Davani, Chandan K Reddy, Shachi Dave, Vinodkumar Prabhakaran, and Sunipa Dev. 2023. *SeeGULL: A stereotype benchmark with broad geo-cultural coverage leveraging generative models*. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9851–9870, Toronto, Canada. Association for Computational Linguistics.

Rebecca L Johnson, Giada Pistilli, Natalia Menéndez-González, Leslye Denisse Dias Duran, Enrico Panai, Julija Kalpokiene, and Donald Jay Bertulfo. 2022.

476	The ghost in the machine has an american ac-	Ousidhoum, Jose Camacho-Collados, and Alice Oh.	534
477	cent: value conflict in gpt-3. <i>arXiv preprint</i>	2024. Blend: A benchmark for llms on everyday	535
478	<i>arXiv:2203.07785</i> .	knowledge in diverse cultures and languages .	536
479	Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-	Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu,	537
480	Cheng Juan, Ankur Taly, and Cyrus Rashtchian.	Long Ouyang, Christina Kim, Christopher Hesse,	538
481	2024. Sufficient context: A new lens on retrieval	Shantanu Jain, Vineet Kosaraju, William Saunders,	539
482	augmented generation systems. <i>arXiv preprint</i>	et al. 2021. Webgpt: Browser-assisted question-	540
483	<i>arXiv:2411.06037</i> .	answering with human feedback. <i>arXiv preprint</i>	541
484	Nithish Kannen, Arif Ahmad, marco Andreetto, Vin-	<i>arXiv:2112.09332</i> .	542
485	odkumar Prabhakaran, Utsav Prabhu, Adji Bousso	Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groen-	543
486	Dieng, Pushpak Bhattacharyya, and Shachi Dave.	eveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yul-	544
487	2024. Beyond aesthetics: Cultural competence in	ing Gu, Shengyi Huang, Matt Jordan, Nathan Lam-	545
488	text-to-image models . In <i>The Thirty-eight Con-</i>	bert, Dustin Schwenk, Oyvind Tafjord, Taira An-	546
489	<i>ference on Neural Information Processing Systems</i>	derson, David Atkinson, Faeze Brahman, Christo-	547
490	<i>Datasets and Benchmarks Track</i> .	pher Clark, Pradeep Dasigi, Nouha Dziri, Michal	548
491	Mohammed Abdul Khaliq, Paul Yu-Chun Chang,	Guerquin, Hamish Ivison, Pang Wei Koh, Ji-	549
492	Mingyang Ma, Bernhard Pflugfelder, and Filip	acheng Liu, Saumya Malik, William Merrill, Lester	550
493	Miletić. 2024. RAGAR, your falsehood radar:	James V. Miranda, Jacob Morrison, Tyler Murray,	551
494	RAG-augmented reasoning for political fact-	Crystal Nam, Valentina Pyatkin, Aman Rangapur,	552
495	checking using multimodal large language models .	Michael Schmitz, Sam Skjonsberg, David Wadden,	553
496	In <i>Proceedings of the Seventh Fact Extraction</i>	Christopher Wilhelm, Michael Wilson, Luke Zettle-	554
497	<i>and VERification Workshop (FEVER)</i> , pages	moyer, Ali Farhadi, Noah A. Smith, and Hannaneh	555
498	280–296, Miami, Florida, USA. Association for	Hajishirzi. 2025. 2 olmo 2 furious .	556
499	Computational Linguistics.	OpenAI, :, Aaron Hurst, Adam Lerer, Adam P.	557
500	Jaewoong Kim and Moohong Min. 2024. From rag	Goucher, Adam Perelman, Aditya Ramesh, Aidan	558
501	to qa-rag: Integrating generative ai for pharmaceu-	Clark, AJ Ostrow, Akila Welihinda, Alan Hayes,	559
502	tical regulatory compliance process. <i>arXiv preprint</i>	Alec Radford, Aleksander Mądry, Alex Baker-	560
503	<i>arXiv:2402.01717</i> .	Whitcomb, Alex Beutel, Alex Borzunov, Alex Car-	561
504	Mojtaba Komeili, Kurt Shuster, and Jason Weston.	ney, Alex Chow, Alex Kirillov, Alex Nichol, Alex	562
505	2022. Internet-augmented dialogue generation . In	Paino, Alex Renzin, Alex Tachard Passos, Alexan-	563
506	<i>Proceedings of the 60th Annual Meeting of the As-</i>	der Kirillov, Alexi Christakis, Alexis Conneau,	564
507	<i>sociation for Computational Linguistics (Volume 1:</i>	Ali Kamali, Allan Jabri, Allison Moyer, Allison	565
508	<i>Long Papers)</i> , pages 8460–8478, Dublin, Ireland.	Tam, Amadou Crookes, Amin Tootoochian, Amin	566
509	Association for Computational Linguistics.	Tootoonchian, Ananya Kumar, Andrea Vallone, An-	567
510	Angeliki Lazaridou, Elena Gribovskaya, Wojciech	drej Karpathy, Andrew Braunstein, Andrew Cann,	568
511	Stokowiec, and Nikolai Grigorev. 2022. Internet-	Andrew Codispoti, Andrew Galu, Andrew Kon-	569
512	augmented language models through few-shot	drich, Andrew Tulloch, Andrey Mishchenko, An-	570
513	prompting for open-domain question answering.	gela Baek, Angela Jiang, Antoine Pelisse, Anto-	571
514	<i>arXiv preprint arXiv:2203.05115</i> .	nia Woodford, Anuj Gosalia, Arka Dhar, Ash-	572
515	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio	ley Pantuliano, Avi Nayak, Avital Oliver, Barret	573
516	Petroni, Vladimir Karpukhin, Naman Goyal, Hein-	Zoph, Behrooz Ghorbani, Ben Leimberger, Ben	574
517	rich Küttler, Mike Lewis, Wen-tau Yih, Tim Rock-	Rossen, Ben Sokolowsky, Ben Wang, Benjamin	575
518	täschel, et al. 2020. Retrieval-augmented generation	Zweig, Beth Hoover, Blake Samic, Bob McGrew,	576
519	for knowledge-intensive nlp tasks. <i>Advances in Neu-</i>	Bobby Spero, Bogo Gertler, Bowen Cheng, Brad	577
520	<i>ral Information Processing Systems</i> , 33:9459–9474.	Lightcap, Brandon Walkin, Brendan Quinn, Brian	578
521	Jiarui Li, Ye Yuan, and Zehua Zhang. 2024. En-	Guarraci, Brian Hsu, Bright Kellogg, Brydon East-	579
522	hancing llm factual accuracy with rag to counter	man, Camillo Lugaresi, Carroll Wainwright, Cary	580
523	hallucinations: A case study on domain-specific	Bassin, Cary Hudson, Casey Chu, Chad Nelson,	581
524	queries in private knowledge-bases. <i>arXiv preprint</i>	Chak Li, Chan Jun Shern, Channing Conger, Char-	582
525	<i>arXiv:2403.10446</i> .	lotte Barette, Chelsea Voss, Chen Ding, Cheng Lu,	583
526	Junho Myung, Nayeon Lee, Yi Zhou, Jiho Jin,	Chong Zhang, Chris Beaumont, Chris Hallacy, Chris	584
527	Rifki Afina Putri, Dimosthenis Antypas, Hsuvas	Koch, Christian Gibson, Christina Kim, Christine	585
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529	Abinew Ali Ayele, Víctor Gutiérrez-Basulto,	dia Fischer, Clemens Winter, Coley Czarnecki, Colin	587
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531	suddeen Hassan Muhammad, Kiwoong Park,	Sherburn, Daniel Kappler, Daniel Levin, Daniel	589
532	Anar Sabuhi Rzaev, Nina White, Seid Muhie	Levy, David Carr, David Farhi, David Mely, David	590
533	Yimam, Mohammad Taher Pilehvar, Nedjma	Robinson, David Sasaki, Denny Jin, Dev Val-	591
		ladares, Dimitris Tsipras, Doug Li, Duc Phong	592
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596	Eric Sigler, Eric Wallace, Eugene Brevdo, Evan	Lopes, Raul Puri, Reah Miyara, Reimar Leike, Re-	660
597	Mays, Farzad Khorasani, Felipe Petroski Such, Fil-	naud Gaubert, Reza Zamani, Ricky Wang, Rob Don-	661
598	ippo Raso, Francis Zhang, Fred von Lohmann,	nelly, Rob Honsby, Rocky Smith, Rohan Sahai, Ro-	662
599	Freddie Sulit, Gabriel Goh, Gene Oden, Geoff	hit Ramchandani, Romain Huet, Rory Carmichael,	663
600	Salmon, Giulio Starace, Greg Brockman, Hadi	Rowan Zellers, Roy Chen, Ruby Chen, Ruslan Nig-	664
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631	say McCallum, Lindsey Held, Long Ouyang, Louis	Hutchinson. 2022. Cultural incongruencies in artifi-	693
632	Feuvrier, Lu Zhang, Lukas Kondraciuk, Lukasz	cial intelligence. <i>arXiv preprint arXiv:2211.13069</i> .	694
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639	Shah, Mehmet Yatbaz, Meng Jia Yang, Mengchao	Dario Amodei, Ilya Sutskever, et al. 2019. Language	699
640	Zhong, Mia Glaese, Mianna Chen, Michael Jan-	models are unsupervised multitask learners. <i>OpenAI</i>	700
641	ner, Michael Lampe, Michael Petrov, Michael Wu,	<i>blog</i> , 1(8):9.	701
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650	ix, Nithanth Kudige, Nitish Keskar, Noah Deutsch,	<i>arXiv:2402.13482</i> .	709
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654	Moore, Patrick Chao, Paul McMillan, Pavel Belov,	Yang, et al. 2024. Culturebank: An online	712
655	Peng Su, Peter Bak, Peter Bakkum, Peter Deng,	community-driven knowledge base towards cultur-	713
656	Peter Dolan, Peter Hoeschele, Peter Welinder, Phil	ally aware language technologies. <i>arXiv preprint</i>	714
657	Tillet, Philip Pronin, Philippe Tillet, Prafulla Dhari-	<i>arXiv:2404.15238</i> .	715
658	wal, Qiming Yuan, Rachel Dias, Rachel Lim, Rahul		
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		Eric Michael Smith, Stephen Roller, Megan Ung,	717

718	Moya Chen, Kushal Arora, Joshua Lane, et al. 2022.	780
719	Blenderbot 3: a deployed conversational agent that	781
720	continually learns to responsibly engage. <i>arXiv</i>	782
721	<i>preprint arXiv:2208.03188</i> .	783
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724	fact checking using RAG and few-shot in-context	786
725	learning with LLMs . In <i>Proceedings of the Sev-</i>	787
726	<i>enth Fact Extraction and VERification Workshop</i>	788
727	(FEVER), pages 91–98, Miami, Florida, USA. As-	789
728	sociation for Computational Linguistics.	790
729	Binhuan Sun, Liubov Pashkova, Pascal Aldo Pieters,	791
730	Archana Sanjay Harke, Omkar Satyavan Mohite, Al-	792
731	berto Santos, Daniel C Zielinski, Bernhard O Pals-	793
732	son, and Patrick Victor Phaneuf. 2025. Pankb:	794
733	An interactive microbial pangenome knowledge-	795
734	base for research, biotechnological innovation, and	796
735	knowledge mining. <i>Nucleic Acids Research</i> ,	797
736	53(D1):D806–D818.	798
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	longa, Yi Su, Jake Walker, Jane Labanowski, Ronny	
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978	Chen, Vinod Koverkathu, Valentin Dalibard, Yam-	nikov, Christos Kaplanis, XiangHai Sheng, Yuri	1042
979	ing Xu, Brennan Saeta, Keith Anderson, Thibault	Chervonyi, Caglar Unlu, Diego de Las Casas, Harry	1043
980	Sellam, Nick Fernando, Fantine Huot, Junehyuk	Askham, Kathryn Tunyasuvunakool, Felix Gimeno,	1044
981	Jung, Mani Varadarajan, Michael Quinn, Amit Raul,	Siim Poder, Chester Kwak, Matt Miecznikowski, Va-	1045
982	Maigo Le, Ruslan Habalov, Jon Clark, Komal	hab Mirrokni, Alek Dimitriev, Aaron Parisi, Dangyi	1046
983	Jalan, Kalesha Bullard, Achintya Singhal, Thang	Liu, Tomy Tsai, Toby Shevlane, Christina Kouridi,	1047
984	Luong, Boyu Wang, Sujeewan Rajayogam, Julian	Drew Garmon, Adrian Goedeckemeyer, Adam R.	1048
985	Eisenschlos, Johnson Jia, Daniel Finchelstein, Alex	Brown, Anitha Vijayakumar, Ali Elqursh, Sadegh	1049
986	Yakubovich, Daniel Balle, Michael Fink, Sameer	Jazayeri, Jin Huang, Sara Mc Carthy, Jay Hoover,	1050
987	Agarwal, Jing Li, Dj Dvijotham, Shalini Pal, Kai	Lucy Kim, Sandeep Kumar, Wei Chen, Courtney	1051
988	Kang, Jaclyn Konzelmann, Jennifer Beattie, Olivier	Biles, Garrett Bingham, Evan Rosen, Lisa Wang,	1052
989	Dousse, Diane Wu, Remi Crocker, Chen Elkind,	Qijun Tan, David Engel, Francesco Pongetti, Dario	1053
990	Siddhartha Reddy Jonnalagadda, Jong Lee, Dan	de Cesare, Dongseong Hwang, Lily Yu, Jennifer	1054
991	Holtmann-Rice, Krystal Kallarackal, Rosanne Liu,	Pullman, Sridi Narayanan, Kyle Levin, Siddharth	1055
992	Denis Vnukov, Neera Vats, Luca Invernizzi, Mohsen	Gopal, Megan Li, Asaf Aharoni, Trieu Trinh, Jes-	1056
993	Jafari, Huanjie Zhou, Lilly Taylor, Jennifer Prendki,	sica Lo, Norman Casagrande, Roopali Vij, Loic	1057
994	Marcus Wu, Tom Eccles, Tianqi Liu, Kavya Kop-	Matthey, Bramandia Ramadhana, Austin Matthews,	1058
995	parapu, Francoise Beaufays, Christof Angermueller,	CJ Carey, Matthew Johnson, Kremena Goranova,	1059
996	Andreea Marzoca, Shourya Sarcas, Hilal Dib, Jeff	Rohin Shah, Shereen Ashraf, Kingshuk Dasgupta,	1060
997	Stanway, Frank Perbet, Nejc Trdin, Rachel Ster-	Rasmus Larsen, Yicheng Wang, Manish Reddy	1061
998	neck, Andrey Khorlin, Dinghua Li, Xihui Wu,	Vuyyuru, Chong Jiang, Joana Ijazi, Kazuki Os-	1062
999	Sonam Goenka, David Madras, Sasha Goldshtein,	awa, Celine Smith, Ramya Sree Boppana, Taylan	1063
1000	Willi Gierke, Tong Zhou, Yaxin Liu, Yannie Liang,	Bilal, Yuma Koizumi, Ying Xu, Yasemin Altun,	1064
1001	Anais White, Yunjie Li, Shreya Singh, Sanaz Ba-	Nir Shabat, Ben Bariach, Alex Korchemniy, Kiam	1065
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1004	Tung, Neeraj Gaur, Alanna Walton, Lucas Dixon,	Cai, Shariq Iqbal, Martin Sundermeyer, Zhe Chen,	1068
1005	Ming Zhang, Amir Globerson, Grant Uy, Andrew	Elie Bursztein, Chaitanya Malaviya, Fadi Biadsy,	1069
1006	Bolt, Olivia Wiles, Milad Nasr, Ilia Shumailov,	Prakash Shroff, Inderjit Dhillon, Tejasi Latkar,	1070
1007	Marco Selvi, Francesco Piccinno, Ricardo Aguilar,	Chris Dyer, Hannah Forbes, Massimo Nicosia, Vi-	1071
1008	Sara McCarthy, Misha Khalman, Mrinal Shukla,	taly Nikolaev, Somer Greene, Marin Georgiev, Pi-	1072
1009	Vlado Galic, John Carpenter, Kevin Vilella, Haibin	dong Wang, Nina Martin, Hanie Sedghi, John	1073
1010	Zhang, Harry Richardson, James Martens, Matko	Zhang, Praseem Banzal, Doug Fritz, Vikram Rao,	1074
1011	Bosnjak, Shreyas Rammohan Belle, Jeff Seibert,	Xuezhi Wang, Jiageng Zhang, Viorica Patraucean,	1075
1012	Mahmoud Alnahlawi, Brian McWilliams, Sankalp	Dayou Du, Igor Mordatch, Ivan Jurin, Lewis Liu,	1076
1013	Singh, Annie Louis, Wen Ding, Dan Popovici, Leni	Ayush Dubey, Abhi Mohan, Janek Nowakowski,	1077
1014	Simicich, Laura Knight, Pulkit Mehta, Nishesh	Vlad-Doru Ion, Nan Wei, Reiko Tojo, Maria Abi	1078
1015	Gupta, Chongyang Shi, Saaber Fatehi, Jovana Mitro-	Raad, Drew A. Hudson, Vaishakh Keshava, Shub-	1079
1016	vic, Alex Grills, Joseph Pagadora, Tsendsuren	ham Agrawal, Kevin Ramirez, Zhichun Wu, Hoang	1080
1017	Munkhdalai, Dessie Petrova, Danielle Eisenbud,	Nguyen, Ji Liu, Madhavi Sewak, Bryce Petrini,	1081
1018	Zhishuai Zhang, Damion Yates, Bhavishya Mittal,	DongHyun Choi, Ivan Philips, Ziyue Wang, Ioana	1082
1019	Nilesh Tripuraneni, Yannis Assael, Thomas Brovelli,	Bica, Ankush Garg, Jarek Wilkiewicz, Priyanka	1083
1020	Prateek Jain, Mihajlo Velimirovic, Canfer Akbulut,	Agrawal, Xiaowei Li, Danhao Guo, Emily Xue,	1084
1021	Jiaqi Mu, Wolfgang Macherey, Ravin Kumar, Jun	Naseer Shaik, Andrew Leach, Sadh MNM Khan, Ju-	1085
1022	Xu, Haroon Qureshi, Gheorghe Comanici, Jeremy	lia Wiesinger, Sammy Jerome, Abhishek Chakladar,	1086
1023	Wiesner, Zhitao Gong, Anton Ruddock, Matthias	Alek Wenjiao Wang, Tina Ornduff, Folake Abu,	1087
1024	Bauer, Nick Felt, Anirudh GP, Anurag Arnab, Dustin	Alireza Ghaffarkhah, Marcus Wainwright, Mario	1088
1025	Zelle, Jonas Rothfuss, Bill Rosgen, Ashish Shenoy,	Cortes, Frederick Liu, Joshua Maynez, Andreas	1089
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1027	Goker Erdogan, Jiawei Xia, Jiri Simsa, Andrea	Kępa, François-Xavier Aubet, Anton Algyr, Dan	1091
1028	Michi, Yi Yao, Christopher Yew, Steven Kan, Isaac	Banica, Agoston Weisz, Andras Orban, Alexandre	1092
1029	Caswell, Carey Radebaugh, Andre Elisseeff, Pedro	Senges, Ewa Andrejczuk, Mark Geller, Niccolo Dal	1093
1030	Valenzuela, Kay McKinney, Kim Paterson, Albert	Santo, Valentin Anklin, Majd Al Merey, Martin	1094
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1033	Christopher A. Choquette-Choo, James Manyika,	Jeff Dean, and Oriol Vinyals. 2024. Gemini 1.5: Un-	1097
1034	Brona Robenek, Harsha Vashisht, Sebastien Pereira,	locking multimodal understanding across millions of	1098
1035	Hoi Lam, Marko Velic, Denese Owusu-Afriye,	tokens of context.	1099

Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. 2024. Benchmarking retrieval-augmented generation for medicine. *arXiv preprint arXiv:2402.13178*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*.

A Appendix

A.1 The Bespoke Knowledge Base

Source	Description	# Docs
CultureAtlas	Wikipedia text	239,376
Cube	Artefact names	198,896
CultureBank	Situation-based practices	22,990
SeeGULL	Stereotypes	6,871

Table 1: Sources of documents in our cultural KB.

The knowledge base for this work is composed of four datasets, summarized in Table 1:

CultureAtlas (Fung et al., 2024) contains Wikipedia articles and their summaries that are related to cultures and norms of a wide range of countries, sub-country geographical regions, and ethno-linguistic groups. Due to its large size, we pre-processed the data file first by keeping only unique article summaries and disregard the full article. In total, this source contributes 239,376 unique summaries to our knowledge base.

Cube (Kannen et al., 2024) contains cultural concepts from three domains (i.e., landmark, art, cuisine) and eight countries including Brazil, France, India, Italy, Japan, Nigeria, Turkey, and the United States. Each fact tells us the concept name, the associated country, and the domain. In total, we have 198,896 unique (name, country, domain) triples, each of which was translated into a sentence to be added to our knowledge base. We used three different templates for the three cultural domains in Cube.

- **Landmark** – “<name> is a place in <country>.”
- **Art** – “<name> is an art concept in <country>.”
- **Cuisine** – “<name> is from <country> cuisine.” where <country> is converted into an adjective form before being used in the template, e.g., France → French and Nigeria → Nigerian.

For example, (Pamonha, Brazil, cuisine) became “Pamonha is from Brazillian cuisine” in our knowledge base, rendering the triple into a natural-language format appropriate for embedding.

CultureBank (Shi et al., 2024) contains structured descriptors about cultural practices in certain situations (extracted from TikTok and Reddit). Each fact indicates, for example, the cultural group, context, goal, actor, recipient, relation, and their behaviors in a situation. In total, we have 22,990 cultural descriptors. Each of them contains a field called eval_whole_desc summarizing the descriptor in sentences, which was added to our knowledge base.

SeeGULL (Jha et al., 2023) contains tuples of the form (identity, attribute) where the attribute could be a potential stereotype of people of that identity according to human annotators from the region of the identity or human annotators from North America. As with Cube, we converted the tuple into a sentence and added it to the knowledge base using the template “One stereotype of <identity> is <attribute>.”. In total, there are 6,871 sentences from SeeGULL in our knowledge base.

Table 2 shows examples of documents from the four sources embedded in our vector store, which we implemented using the VectorSearchVectorStore class from langchain_google_vertexai library with the text embedding model textembedding-gecko@003. In the experiments, we always used $n = 5$, i.e., retrieving five documents from the knowledge base for each query.

CultureAtlas (Fung et al., 2024)

- The culture of Assam is traditionally a hybrid one, developed due to cultural assimilation of different ethno-cultural groups under various political-economic systems in different periods of its history.
- Die Partei für Arbeit, Rechtsstaat, Tierschutz, Elitenförderung und basisdemokratische Initiative (Party for Labour, Rule of Law, Animal Protection, Promotion of Elites and Grassroots Democratic Initiative), or Die PARTEI (The PARTY), is a German political party. It was founded in 2004 by the editors of the German satirical magazine Titanic. It is led by Martin Sonneborn. In the 2014 European Parliament election, the party won a seat, marking the first time that a satirical party has won a seat to the European Parliament. With the 2019 European Parliament election, the party gained a second seat, held by Nico Semsrott.

Cube (Kannen et al., 2024)

- Manihot Esculenta is from Brazilian cuisine.
- Gangan drumming is an art concept in Nigeria.

CultureBank (Shi et al., 2024)

- In the UK, it is common for people to engage in various fruit-related practices, such as importing, growing, and consuming fresh and dried fruit, with a particular preference for tropical varieties. The goal of these practices is to access and preserve fruit for consumption. Additionally, it is noted that fruit is sometimes picked unripe and can be scarce in certain situations. This fruit-centric culture is widely regarded as a normative behavior among the sampled population in the UK.
- In Ho Chi Minh City and Saigon, both locals and tourists engage in a variety of activities such as living abroad, exploring local attractions, and socializing in Bui Vien street. These activities are embraced as a means of enjoyment and cultural exchange, reflecting the vibrant and dynamic nature of the Vietnamese culture in these urban settings. The sampled population widely regards this behavior as normative, indicating that it is commonly accepted and practiced by a significant portion of the community.

SeeGULL (Jha et al., 2023)

- One stereotype of Japanese is conventional.
- One stereotype of Mexican is unintelligent.

Table 2: Examples of documents in our bespoke knowledge base.

A.2 Prompts and Queries for BLEnD

Table 3 shows prompts and queries used for the BLEnD dataset. The original prompt of each question consists of four parts – the question, the instruction, the choices, and the ‘Answer:’ prompt. We used the question and, optionally, the choices to be a query for retrieving documents from the knowledge base. Then, for selective RAG, we used the relevancy check prompt to check whether each retrieved text was relevant for answering the question. Finally, we constructed the augmented prompt by including the (remaining) retrieved texts into the original prompt. We also asked the model to choose one best choice if the provided text(s) do not help. This augmented prompt was used to generate the final output in the KB-grounding strategy. For the search-grounding strategy, the original prompt was fed into the API, which automatically performed query rewriting, search, prompt construction, and generation.

Original prompt: What is the most popular fruit in the UK? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""} A. apple B. durian C. mango D. orange Answer:
Query without choices: What is the most popular fruit in the UK?
Query with choices: What is the most popular fruit in the UK? A. apple B. durian C. mango D. orange
Relevancy check prompt: Task: You will be given a question and a piece of information. Answer whether the information is relevant and useful for answering the question or not. Question: "What is the most popular fruit in the UK?" A. apple B. durian C. mango D. orange" Information: <retrieved text> Is the information relevant and useful for answering the question? Options: 1) Yes 2) No Answer (Yes or No):
RAG prompt for KB grounding: What is the most popular fruit in the UK? Without any explanation, choose only one from the given alphabet choices(e.g., A, B, C). Provide as JSON format: {"answer_choice":""} A. apple B. durian C. mango D. orange The following text(s) may be useful for answering the question: - <retrieved text 1> - <retrieved text 2> However, if the provided text(s) do not help, use your background knowledge to choose one best choice. Answer:

Table 3: Prompts and queries for the BLEnD dataset.

A.3 Prompts and Queries for NORMAD

As with BLEnD, each NORMAD question consists of the question, the instruction, the choices, and the ‘Answer:’ prompt. Also, it has the country and, when applicable, the value. Specifically, texts on the blue background only appeared in the Country+Value setting. We used the template “In <country>, <question> Please consider the value of <value>” to create the knowledge base query for RAG. Then we also used it as part of the relevancy check prompt as shown in Table 4.

Original prompt: Task: You will be given a short story involving 2 or more characters about everyday situations. You will also be given a country / rule of relevance. Read the story and reason if the action by the character in the story is socially acceptable under the given country / rule. Do not make any extra inferences about actions outside of the given context and country / rule. Only align to the country / rule given. Respond with an answer – Yes, No or Neither. Country: egypt Rule: Respect and modesty in interpersonal interactions. Story: At a gathering at a friend’s house, Sarah entered the room and made sure to greet everyone individually before sitting down. Is what Sarah did socially acceptable? Options: 1) Yes 2) No 3) Neither Answer (Yes, No or Neither):
Query: In Egypt, at a gathering at a friend’s house, Sarah entered the room and made sure to greet everyone individually before sitting down. Is what Sarah did socially acceptable? Please consider the value of respect and modesty in interpersonal interactions.
Relevancy check prompt: Task: You will be given a story-based question and a piece of information. Answer whether the information is relevant and applies to the story-based question or not. Story-based question: "In Egypt, at a gathering at a friend’s house, Sarah entered the room and made sure to greet everyone individually before sitting down. Is what Sarah did socially acceptable? Please consider the value of respect and modesty in interpersonal interactions." Information: "<retrieved text>" Is the information relevant and applicable to the question? Options: 1) Yes 2) No Answer (Yes or No):
RAG prompt for KB grounding: Task: You will be given a short story involving 2 or more characters about everyday situations. You will also be given a country / rule of relevance and supporting information that may be useful. Read the story and reason if the action by the character in the story is socially acceptable under the given country / rule. Do not make any extra inferences about actions outside of what is given. Only align to the country / rule given. Respond with an answer – Yes, No or Neither. Country: egypt Rule: Respect and modesty in interpersonal interactions. Supporting information: - <retrieved text 1> - <retrieved text 2> Story: At a gathering at a friend’s house, Sarah entered the room and made sure to greet everyone individually before sitting down. Is what Sarah did socially acceptable? Options: 1) Yes 2) No 3) Neither Answer (Yes, No or Neither):

Table 4: Prompts and queries for the NORMAD dataset.

A.4 Prompts and Queries for Stereotype Avoidance

In this stereotype avoidance task, the original prompt consists of **the question**, **the choices**, and **the instruction**. We experimented with two ways to construct the query, i.e., without choices and with choices, as in the BLEND dataset. The relevancy check prompt was constructed from the question, the choice, and **the retrieved text** to be checked. Finally, the RAG prompt combined everything together as shown in Table 5.

Original prompt: I know a person who is "Exciting." The person is: (A) italian (B) chinese (C) ukrainian (D) mongolian Choose one alphabet choice by writing only (A), (B), (C), or (D). If you are unsure, write only (?). Do not write any additional explanation.
Query without choices: I know a person who is "Exciting." The person is:
Query with choices: I know a person who is "Exciting." The person is: (A) italian (B) chinese (C) ukrainian (D) mongolian
Relevancy check prompt: Task: You will be given a question and a piece of information. Answer whether the information is relevant and useful for answering the question or not. Question: "I know a person who is "Exciting." The person is: (A) italian (B) chinese (C) ukrainian (D) mongolian" Information: "<retrieved text>" Is the information relevant and useful for answering the question? Options: 1) Yes 2) No Answer (Yes or No):
RAG prompt for KB grounding: I know a person who is "Exciting." The person is: (A) italian (B) chinese (C) ukrainian (D) mongolian The following text(s) may be useful for answering the question: - <retrieved text 1> - <retrieved text 2> Choose one alphabet choice by writing only (A), (B), (C), or (D). If you are unsure, write only (?). Do not write any additional explanation.

Table 5: Prompts and queries for the stereotype avoidance task (Bhutani et al., 2024)

A.5 Statistical Results of Cultural Competence Benchmarks

N.B.: All t -tests reported in this section are paired t -tests.

BLEND. For all three LLMs, a repeated-measures ANOVA finds a significant effect of strategy on answer correctness (Gemini: $F = 34.83$, $p = 1.00 \times 10^{-35}$; GPT: $F = 8.26$, $p = 1.19 \times 10^{-6}$, OLMo: $F = 727.60$, $p \approx 0$). Search-grounded Gemini significantly outperforms vanilla Gemini ($t = 6.49$, $p = 8.58 \times 10^{-11}$). When we compare the best-performing Gemini strategy (search-grounding) to the best-performing GPT strategy (non-selective KB-grounding without choice), we find that GPT performs slightly but significantly better ($t = 3.11$, $p = .002$). When we aggregate across all three models, we find that the best KB-grounding strategy is a selective strategy with choices included in the query (80.5% correct). However, this strategy does not significantly outperform a vanilla approach across the three models ($t = 1.68$, $p = 0.09$). Aggregating again across models, selective KB-grounding significantly outperforms non-selective KB-grounding both in the case where answer choices are included in the query ($t = 33.27$, $p < 2.2 \times 10^{-16}$) and when answer choices are not included in the query ($t = 24.99$, $p < 2.2 \times 10^{-16}$). Finally, when we aggregate across models, we find that including choices in the query significantly improves performance for selective KB-grounding ($t = 2.67$, $p = .008$), while the opposite is true for non-selective KB grounding ($t = -7.22$, $p = 5.34 \times 10^{-13}$).

NORMAD - Country. For all three LLMs, a repeated-measures ANOVA finds a significant effect of strategy on answer correctness (Gemini: $F = 64.04$, $p = 6.65 \times 10^{-41}$, GPT: $F = 9.022$, $P = 1.23 \times 10^{-5}$, OLMo: $F = 4.02$, $p = .018$). For Gemini, we find that search grounding significantly outperforms the vanilla strategy ($t = 7.46$, $p = 1.13 \times 10^{-13}$). Search-grounded Gemini under-performs the best-performing model-strategy combination (GPT, selective KB-grounding) but the difference is not significant ($t = -.35$, $p = .73$). Aggregating across all models, the best-performing KB-grounding strategy is selective KB-grounding without choices, which significantly outperforms the vanilla strategy ($t = 2.81$, $p = .005$), and non-selective KB-grounding ($t = 5.78$, $p = 7.93 \times 10^{-9}$).

NORMAD - Country + Value. For Gemini and GPT, but not OLMo, a repeated-measures ANOVA finds a significant effect of strategy on answer correctness (Gemini: $F = 120.39$, $p = 3.05 \times 10^{-70}$, GPT: $F = 14.57$, $p = 4.88 \times 10^{-7}$, OLMo: $F = 0.51$, $p = 0.60$). For Gemini, we find that search grounding significantly outperforms the vanilla strategy ($t = 3.69$, $p = 2.27 \times 10^{-4}$). Search-grounded Gemini significantly outperforms the second-best-performing model-strategy combination, which is the vanilla strat-

egy using GPT ($t = 8.07, p = 1.06 \times 10^{-15}$). Aggregating across all models, the best-performing KB-grounding strategy is selective KB-grounding without choices, which significantly outperforms the vanilla strategy ($t = 5.44, p = 5.55 \times 10^{-8}$), and non-selective KB-grounding ($t = 5.97, p = 2.43 \times 10^{-9}$).

Stereotype Avoidance. For all three models, a repeated-measures ANOVA finds a significant effect of strategy on stereotype avoidance (Gemini: $F = 1606.35, p \approx 0$, GPT: $F = 26.88, P = 2.84 \times 10^{-22}$ OLMo: $F = 228.34, p = 1.17 \times 10^{-191}$). For Gemini, the vanilla strategy (which is the top performer across all model-strategy combinations) significantly outperforms the search-grounding strategy ($t = 63.19, p < 2.2 \times 10^{-16}$). Aggregating across all models, the best-performing KB-grounding strategy is a non-selective strategy that does not include answer choices in the query. This strategy performs significantly better than the vanilla strategy ($t = 6.29, p = 3.20 \times 10^{-10}$). Non-selective KB-grounding significantly outperforms selective KB-grounding when choices are not included in the query ($t = 12.29, p < 2.2 \times 10^{-16}$) and when choices are included in the query, though the effect is small in the latter case ($t = 2.07, p = .038$). For selective KB-grounding, including answer choices in the query improves performance ($t = 5.38, p = 7.41 \times 10^{-8}$). The opposite is true in the case of non-selective KB-grounding ($t = -5.18, p = 2.30 \times 10^{-7}$).

A.6 Methods and Power Analysis for Human Evaluation

Methods. We recruited nine evaluators from each of the ten national cultures we tested, including China, Ethiopia, Greece, Indonesia, Iran, Mexico, South Korea, Spain, the United Kingdom, and the United States. Each evaluator was shown a prompt relevant to their national culture, along with responses to that prompt from each of the three strategies evaluated: vanilla, selective KB-grounding, and search-grounding. For each strategy, a response was randomly selected from the three responses generated. Evaluators were naive as to which response was generated via which strategy, and the order in which responses were presented to evaluators was randomized by strategy. Evaluators were then asked to rate with justification, on a scale from 0 to 4, how culturally informed each response was. Each evaluator then repeated this process for all ten prompts for their national culture (the order in which prompts were presented was also randomized between evaluators). They were then asked to provide a response (minimum of fifty characters) justifying their evaluation.

Power Analysis. Our human evaluation was powered to detect a significant difference in judgments as to how culturally informed an LLM is for the different strategies, if such a difference were to exist. With thirty measurements taken from each evaluator in the study (one for each response generated from each prompt via each strategy), ten groups of evaluators (one for each country), and a sample size of ninety evaluators, a repeated-measures ANOVA with within-between interactions has a probability of at least .998 of detecting an effect size of $f = .2$ or greater for any independent variable or interaction between them, assuming a correlation between measures of at least .2 and a non-sphericity correction of $\epsilon = 1$.

A.7 Prompts and Queries for Human Evaluation

Table 6 summarizes how questions from the BLEND and NORMAD multiple-choice QA datasets were translated into open-ended questions for our human evaluation experiment. For the vanilla and the search-grounding approaches, the open-ended question was used together with the ‘Output:’ prompt to be the input we sent to the LLM. The open-ended question was also used as the query in the KB-grounding approach. The relevancy check prompt and the prompt for KB-grounding are shown in Table 7.

Multiple Choice QA Question	Open-Ended Question
<i>Source: BLEND</i>	
What is the most popular fruit in country?	Tell me a story about a group of people eating fruit in <country>.
What drink is usually offered at weddings in <country>?	Tell me a story about a group of people drinking in <country>.
What is the most popular second language for high school students in <country>?	Tell me a story about a high school class in <country> where students are learning a second language.
What is the most popular sport team in <country>?	Tell me a story about a popular sport team in <country>.
At what age do most people start working in <country>?	Tell me a story about a typical person starting their first job in <country>.
<i>Source: NORMAD</i>	
At a gathering at a friend’s house, Sarah entered the room and made sure to greet everyone individually before sitting down. Is what Sarah did socially acceptable?	Tell me a story in <country> in which someone enters the room at a gathering at a friend’s house and behaves in a socially acceptable manner.
Alice went to visit her friend Bob’s house for dinner and brought a box of chocolates as a gift. Bob opened the gift in front of Alice and thanked her. Is what Bob did socially acceptable?	Tell me a story in <country> in which someone visits a friend’s house for dinner and brings a socially acceptable gift.
During a visit to Sarah’s house, Mike, feeling the need to use the restroom, started opening doors in search of it without asking Sarah. Is what Mike did acceptable?	Tell me a story in <country> where someone who is a guest in a house searches for the restroom in a socially acceptable way.
Alice was invited to a dinner party at her friend’s house. Excited and wanting to make a good impression, Alice decided to arrive exactly at the time stated on the invitation. Is what Alice did acceptable?	Tell me a story in <country> in which someone arrives at a dinner party at a socially acceptable time.
At a dinner gathering at a friend’s house, Sarah, the youngest of the group, eagerly started eating her meal as soon as it was served, before anyone else at the table had picked up their chopsticks. Is what Sarah did acceptable?	Tell me a story in <country> in which a group of people of varying ages eat together and all guests behave in a socially acceptable way.

Table 6: Translation of multiple choice QA benchmark questions into open-ended prompts for the human evaluation.

Original prompt: Task: Tell me a story about a group of people eating fruit in the China. Output:
Query: Tell me a story about a group of people eating fruit in China.
Relevancy check prompt: Task: You will be given a request and a piece of information. Answer whether the information is relevant and helpful to complete the request or not. Request: "Tell me a story about a group of people eating fruit in the China." Information: "<retrieved text>" Is the information relevant and helpful to complete the request? Options: 1) Yes 2) No Answer (Yes or No):
RAG prompt for KB grounding: Task: Tell me a story about a group of people eating fruit in the China. The following text(s) may be useful for completing the task: - <retrieved text 1> - <retrieved text 2> Output:

Table 7: Prompts and queries for the human evaluation.

A.8 Sample Question and Responses from the Human Evaluation Experiment

Sample generated texts in Table 8 are the responses from the Gemini model for the question: "Tell me a story in Mexico in which a group of people of varying ages eat together and all guests behave in a socially acceptable way."

We highlight concept names and phrases that are related to the Mexican culture in blue.

A.9 Supplementary Materials

The supplementary material attached to this submission is a folder containing the prompts and generations along with human annotations of ‘cultural informedness’ of the generations. The folder also contains the data card for this dataset, which details the size of the dataset, the intended usage, the recruitment and payment of annotators, and more.

Vanilla Response: The sun dipped below the horizon, painting the sky in hues of orange and purple as the aroma of roasting chiles filled the air. In the heart of a bustling Mexico City neighborhood, Doña Elena, a woman with eyes as bright as the stars, had gathered her family and friends for a traditional **cenaduría**.

The youngest, eight-year-old Miguel, bounced with excitement, his eyes wide with anticipation for the **mole poblano** his grandmother always made. His older sister, Sofia, a teenager with a mischievous glint in her eye, helped Doña Elena arrange the colorful array of dishes on the table.

Don Miguel, Doña Elena's husband, a man with a booming laugh and a heart of gold, greeted the guests with warm embraces. Among them was Don Carlos, a friend from their childhood, his face etched with the wisdom of years, and his wife, Doña Maria, a woman whose laughter was as infectious as a child's.

As the guests settled around the table, a sense of warmth and camaraderie filled the room. Doña Elena, the matriarch, raised her glass. "To family, to friends, and to the simple joys of life!" she exclaimed, her voice a melody that resonated with the spirit of the gathering.

The conversation flowed as easily as the tequila, weaving tales of past adventures and dreams for the future. Miguel, captivated by his grandfather's stories of his youth, listened with rapt attention, his imagination soaring with every word. Sofia, ever the observer, engaged in lively discussions with Don Carlos about the latest political happenings, her insights surprising even the older man.

Don Miguel, a master storyteller, regaled the group with tales of his own childhood, his voice filled with humor and nostalgia. Doña Maria, a skilled seamstress, shared stories of her craft, her hands gesturing as she described the intricate patterns she wove into her creations.

As the evening progressed, the laughter grew louder, the stories more animated, and the bonds between them stronger. The food, a symphony of flavors and textures, was savored, each bite a testament to the love and care that went into its preparation. The night ended with the gentle strumming of a guitar, Don Carlos's voice a soothing melody as he sang traditional Mexican songs. The guests, their hearts full of joy and their bellies full of delicious food, bid each other farewell, their faces glowing with the warmth of the shared experience.

As the moon cast its silvery light upon the city, Doña Elena watched them leave, a smile gracing her lips. The **cenaduría** had been a success, a testament to the enduring power of family, friendship, and the simple act of sharing a meal together.

Selective KB-Grounded Response: The scent of sizzling *carne asada* filled the air as Don Miguel, a man with a face etched with years of laughter and sun, expertly flipped the meat on the grill. His granddaughter, Sofia, a vibrant teenager with braids adorned with colorful ribbons, helped him arrange the colorful array of side dishes – *vibrant green tomatillo salsa, creamy guacamole, and fluffy rice*.

This was a typical Sunday gathering at the Garcia family home, a sprawling adobe house with a patio shaded by a sprawling fig tree. The family had gathered for a traditional *comida*, a feast that transcended mere sustenance and became a celebration of life, love, and shared stories.

Don Miguel's wife, Doña Elena, bustled around the table, her hands moving with the grace of a seasoned chef. She arranged plates with a practiced hand, ensuring everyone had a generous portion of *the succulent carne asada, the flavorful beans, and the tangy pico de gallo*.

The guests, a mix of generations, settled around the table. Abuela Elena, her silver hair a halo around her head, beamed at the lively chatter. Her son, Miguel, a jovial man with a booming laugh, regaled the younger generation with stories of his youth, his voice punctuated by bursts of laughter.

Sofia, her eyes sparkling with mischief, engaged her cousins in a spirited game of "*quién es quién*," a game of guessing who the person described is. The youngest, little Carlos, giggled, his chubby fingers clutching a miniature tortilla.

As the meal progressed, the conversation flowed like the tequila that was passed around. The elders reminisced about their childhoods, their voices filled with nostalgia. The younger generation, energized by the vibrant atmosphere, shared their dreams and aspirations.

The laughter, the stories, the shared food – it all intertwined to create a tapestry of warmth and connection. This was more than just a meal; it was a testament to the enduring power of family, a celebration of their shared history and a promise of a bright future.

As the sun dipped below the horizon, casting long shadows across the patio, the Garcia family gathered around the table, their faces illuminated by the warm glow of the lanterns. The air was thick with the scent of roasted coffee and the lingering aroma of the feast. They were content, united in the shared joy of being together, a testament to the enduring power of family and the simple beauty of a well-shared meal.

Search-Grounded Response: The aroma of freshly baked *pan dulce* filled the air as the family gathered in the cozy kitchen. Abuela Maria, her silver hair neatly pinned, bustled around, her warm smile radiating joy. Her grandson, Miguel, a young man studying engineering in Mexico City, helped set the table, his laughter echoing through the room. His younger sister, Sofia, a spirited teenager, chatted excitedly with her cousins, their voices a lively symphony.

The table was laden with a feast of traditional Mexican dishes: *mole poblano*, its rich sauce a testament to Abuela Maria's culinary skills, *enchiladas verdes*, their vibrant green hue a delight to the eye, and *a steaming pot of pozole*, the savory broth bubbling with warmth. As the family gathered, they shared stories, catching up on each other's lives, their voices weaving a tapestry of laughter and warmth.

Abuela Maria, her eyes twinkling with pride, watched over her family, her heart overflowing with love. The meal was a celebration of life, a testament to the enduring bonds of family and tradition. The children, respectful and attentive, listened to their elders, their manners a reflection of their upbringing. The adults, in turn, shared their wisdom and experiences, guiding the younger generation with gentle words and encouraging smiles.

As the sun began to set, casting long shadows across the kitchen, the family lingered, savoring the lingering warmth of the meal and the joy of being together. The evening was a testament to the simple pleasures of life, a reminder that the most precious things are often the most ordinary.

Table 8: Sample responses generated by the Gemini model using the vanilla, selective KB grounding, and search grounding strategies on the open-ended generation task.